Hybrid Convolution Reparameterization for Efficient Deep Learning-Based Nonprecipitation Echo Recognition and Removal

Jianwei Si, Lei Han, *Member, IEEE*, Lejian Zhang, and Chuanxin Li

***Abstract*—Weather radar reflectivity images play a critical role in reliable weather monitoring and forecasting. However, inherent factors such as ground clutter, sea clutter, and electromagnetic interference frequently introduce nonprecipitation echoes (NPEs) into these data, posing significant challenges for accurate precipitation detection. A promising solution is leveraging deep learning networks to identify and remove NPEs using satellite observations. To enhance the practicality of these models, recent advancements in reparameterization technology have shown potential for reducing computational complexity. However, existing methods focus on parallel multibranch reparameterization; the fusion of multiple convolutions into single equivalent convolutions, referred to as multiconvolution reparameterization, is still not explored. In this study, we propose a novel reparameterized NPE removal network (RepNPE-Net), designed to recognize and remove NPEs of radar reflectivity data using multichannel brightness temperature (BT) observations from a geostationary meteorological satellite. Reparameterized NPE removal network (RepNPE)-Net incorporates two innovative dual-stream convolution structure-based modules, including the reparameterized dual-stream convolutional module (RepDCM) and the reparameterized attention dual-stream convolutional module (RepADCM), which synergize standard and depthwise separable (DS) residual convolutional blocks to improve feature extraction and representation capabilities. Within the RepADCM module, a positional efficient local attention (PELA) block is designed to enable the network to focus on spatially significant positional features and enhance model's accuracy. Furthermore, to strengthen the practical application ability of the proposed RepNPE-Net, we introduce hybrid convolution reparameterization (HCR) technology, which consolidates multibranch and multiconvolution operations (e.g., depthwise (DW) and pointwise (PW) convolutions) into single equivalent convolutions during the inference stage, significantly reducing computational complexity without compromising network performance. Experimental results demonstrate that RepNPE-Net outperforms existing methods in both NPE removal accuracy and computational efficiency, highlighting its potential for improving radar data quality and advancing meteorological research and applications.**

***Index Terms*—Deep learning, geostationary meteorological satellite, nonprecipitation echo recognition and removal, reparameterization.**

I. Introduction

WEATHER radar reflectivity data are essential for monitoring atmospheric phenomena, particularly in precipitation detection and quantification. However, these data often contain nonprecipitation echoes (NPEs) due to ground and sea clutter, anomalous propagation, electromagnetic interference, and so on. Although these signals do not represent actual precipitation, they are difficult to distinguish from genuine precipitation echoes, significantly degrading the accuracy of radar-derived applications such as quantitative precipitation estimation, radar extrapolation nowcasting, and radar reflectivity retrieval [1], [2]. Consequently, accurate identification and removal of these NPEs have become a critical focus in radar meteorology research.

Existing multisource data-based NPE removal methods predominantly treat geostationary satellite observations and rain gauge data as auxiliary data sources to enhance NPE recognition and elimination. In particular, satellite brightness temperature (BT) data provide valuable cloud-top atmospheric information that supports NPE identification [3]. For example, Guo et al. [3] adopted BT and rain gauge data as inputs to a fuzzy logic algorithm to remove NPEs. Cheng et al. [4] employed BT observations and a threshold method to identify and eliminate NPEs. Despite achieving improved advancements, the above methods still face two major limitations. One is the over-reliance on simplified statistical models and the insufficient consideration of spatial and contextual information. These methods often assume linear or simplified relationships between satellite observations and precipitation, ignoring complex nonlinear interactions and spatial patterns, leading to inaccurate classification and elimination results. Another is the indirect utilization of auxiliary data. Satellite and rain gauge data are generally transformed into thresholds, probability statistics, or auxiliary indicators, limiting the sufficient exploitation of their spatiotemporal information and reducing the effectiveness of classification and elimination.

Recently, deep learning technology has demonstrated strong capabilities in automatically learning complex spatial and temporal patterns from large-scale data, offering a promising

Received 18 January 2025; revised 17 October 2025 and 31 January 2026; accepted 16 March 2026. Date of publication 7 April 2026; date of current version 2 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42275003 and in part by the National Key Research and Development Program of China under Grant 2025YFE0217100. *(Corresponding authors: Lei Han; Lejian Zhang.)*

Jianwei Si and Lei Han are with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao 266404, China, and also with China Meteorological Administration Tornado Key Laboratory, Foshan 528000, China (e-mail: sjw9002@stu.ouc.edu.cn; hanlei@ouc.edu.cn).

Lejian Zhang is with the Meteorological Observation Center, China Meteorological Administration, Beijing 100081, China (e-mail: zhanglejian@cma.gov.cn).

Chuanxin Li is with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao 266404, China (e-mail: lichuanxin@stu.ouc.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2026.3675666

solution to the aforementioned limitations. By directly using satellite images as inputs and rain gauge data as outputs, these models can fully exploit the rich information provided by these auxiliary data sources and improve NPE classification and elimination. However, the sparsity of rain gauge stations necessitates a point-based classification design (e.g., rainfall or nonrainfall points) in these networks, which introduces considerable computational overhead in large-scale applications.

To address this computational challenge, several studies have explored reparameterization techniques in deep learning networks. Ding et al. [5] proposed to reparameterize 1-D asymmetric convolutions into standard square convolutions to enhance representation capacity. Ding et al. [6] and Wang et al. [7] merged multibranch 3×3 and 1×1 convolutions with residual connections into single 3×3 convolutions for efficient inference. Ding et al. [8] combined branches of varying scales and complexities into a single block for inference. Guo et al. [9] and Ding et al. [10] reparameterized multibranch dilated convolutions (i.e., 3×3, 5×5, and 9×9 convolutions) into a single large-kernel convolution. Luo et al. [11] and Wu et al. [12] merged multibranch convolutions into single layers for improved inference efficiency or support incremental learning. Vasu et al. [13] reparameterized each branch of DW and PW convolutions before merging them into a single depthwise separable (DS) convolution for efficient inference. All these methods adopt parallel multibranch reparameterization to streamline inference. However, multiconvolution reparameterization, which fuses multiple hierarchical convolutions (e.g., DW and PW convolutions in DS structure) into single equivalent convolutions, remains unexplored in deep learning reparameterization studies, leading to higher inference time.

In this study, we propose a reparameterized convolutional neural network (CNN)-based NPE removal network named RepNPE-Net, designed to utilize multichannel BT data of Fengyun-4A (FY-4A) meteorological satellite to identify and remove NPEs of radar reflectivity data. In particular, two reparameterized modules based on dual-stream convolution structure, including reparameterized dual-stream convolutional module (RepDCM) and reparameterized attention DCM (RepADCM), are introduced. These modules integrate standard convolution and DS convolution, enabling the network to capture complex feature representations from satellite data. In the RepADCM module, a positional embedding-based efficient local attention (PELA) block is designed, allowing the model to selectively focus on spatially important positional features and effectively enhancing the model's representation capability. Additionally, we develop a novel hybrid convolution reparameterization (HCR) technology for DS residual convolutional block, which reparameterizes multibranch and multiconvolutional operations into single equivalent convolutions during the inference stage, significantly reducing computational complexity while preserving network's representation capacity.

The rest of this article is organized as follows. Section II reviews the related works of NPE recognition and removal, as well as recent advances in reparameterization techniques. Section III introduces the data used in this study. Section IV presents the architecture and specific details of the proposed RepNPE-Net. The performance evaluation of the proposed network and other comparison methods is detailed in Section V. Finally, Section VI encapsulates the conclusion of this study.

II. Related Works

In our study, we focus on NPE recognition and removal leveraging reparameterization technology to improve computational efficiency while preserving representation capacity. To contextualize our contributions, we review related works from two key perspectives: 1) existing methods for NPE and removal and 2) advances in reparameterization techniques across different tasks.

A. Nonprecipitation Echo Recognition and Removal

Existing NPE removal methods can be broadly categorized into two types: radar-based approaches and multisource data-based approaches. The former primarily relies on radar-intrinsic parameters such as reflectivity, correlation coefficients, radial velocity, and polarimetric measurements to identify and eliminate NPEs. Steiner and Smith [14] proposed an automated procedure that utilizes the 3-D structure of radar reflectivity to identify and eliminate NPEs. Cho et al. [15] proposed a fuzzy logic-based algorithm that employs radial velocity, vertical reflectivity gradient, and reflectivity to remove NPEs, effectively eliminating ground clutter and anomalous propagation echoes. Tang et al. [16] developed a machine learning-based method for NPE removal, where the radar reflectivity and environmental conditions are input into an automated classifier to recognize and eliminate NPEs. Gourley et al. [17] proposed a polarimetric fuzzy logic-based algorithm that adopts five radar parameters and a fuzzy logic algorithm to identify and eliminate NPEs. Krause [18] proposed the MetSignal removal algorithm, where dual-polarization radar data are regarded as input features and fed into a fuzzy logic network to distinguish and remove NPEs. Lakshmanan et al. [19] developed a neural network-based removal algorithm to recognize and eliminate NPEs from radar data. The algorithm utilizes polarimetric parameters such as reflectivity, Doppler moments, and 3-D volume features to compute pixel-wise precipitation probabilities, which are subsequently clustered and averaged to produce the final classification results for NPE removal. Lee et al. [20] proposed a fuzzy inference system-based algorithm to remove line-shaped NPEs using reflectivity data, sun strobe characteristics, and radial interference echo features. Tahanout and Chatelet [21] introduced the filtering by coherence (FILCOH) technique to mitigate NPEs in radar data by leveraging differences in correlation time, showing strong performance across multiple pulse repetition time schemes. Additionally, Lepetit et al. [22] proposed a CNN-based method that utilizes polarimetric radar parameters as input, which are then fed into the widely used U-Net [23] architecture to identify and remove NPEs. Despite significant developments in the aforementioned NPE removal methods, their performance remains constrained by the inherent poor quality of radar data.

To address the issue, some researchers have explored multisource data such as geostationary satellite images and rain gauge data to enhance the accuracy of NPE removal missions. Guo et al. [3] proposed a fuzzy logic-based removal algorithm, which combines satellite data, radar observations, and gauge data to identify and remove NPEs, demonstrating improvements over traditional threshold-based methods. Cheng et al. [4] proposed a multisource data-based removal algorithm that utilizes BT observations from Himawari-8 meteorological satellite along with a threshold method to identify and eliminate NPEs. These studies demonstrate the potential of multisource data for improving NPE removal, motivating us to directly integrate satellite and rain gauge data within deep learning frameworks, as explored in this study.

B. Reparameterization Technology

Recently, advancements in deep learning have highlighted reparameterization technology as an effective method to optimize network structures and enhance inference efficiency. Ding et al. [5] proposed the asymmetric convolutional network (ACNet), which reparameterizes 1-D asymmetric convolutions into standard square convolutions to enhance representation capacity and reduce inference time. Ding et al. [6] introduced a reparameterization approach that integrates multibranch structures incorporating 3×3 and 1×1 convolutions and residual connections during training, which are then merged into single 3×3 convolutions for inference, significantly improving deployment efficiency. Similarly, Wang et al. [7] applies reparameterization to merge multibranch convolutional and attention modules into single convolutions during inference, achieving efficient computation while maintaining performance. Ding et al. [8] proposed the diverse branch block (DBB), which combines branches of varying scales and complexities during training, and reparameterizes them into single convolution for efficient inference. Guo et al. [9] proposed a reparameterized large-kernel convolutional network for weed detection, where dilated reparam blocks transform multiple dilated layers into equivalent single large-kernel convolutions during inference, effectively enhancing detection performance. Similarly, Ding et al. [10] leverage reparameterization to merge multibranch dilated convolutions into single large-kernel convolutions, improving inference efficiency while preserving feature representation. Luo et al. [11] developed a novel localized and layered reparameterization, which progressively merges multibranch parameters across layers into compact single-branch convolutions, enhancing efficiency while supporting incremental learning. Wu et al. [12] proposed an expert-like reparameterization of heterogeneous pyramid receptive fields. Heterogeneous three-scale convolutions (i.e., 1×3, 3×1, and 3×3 convolution) are merged into a single 3×3 convolution, such as five- and seven-scale convolutions. These reparameterized convolutions are further converted into a single 7×7 convolution to enhance medical image classification performance. Similar to [12], Guo et al. [24] proposed a reparameterization CNN for infrared solar panel fault classification, where heterogeneous convolutions are merged into a single convolution for inference. Ding et al. [25] introduced a structural reparameterization method that integrates local priors into a fully connected (FC) layer via merging the trained parameters of a parallel conv kernel into the FC kernel, effectively balancing computational efficiency with model accuracy. Ding et al. [26] proposed a structural reparameterization method for lossless channel pruning. During training, convolutional layers are divided into "remembering" and "forgetting" parts, which are later merged into a slimmer architecture for inference, significantly reducing computational costs without sacrificing accuracy.

Additionally, combining reparameterization with DS convolutions, Vasu et al. [13] introduced a lightweight backbone for mobile devices, where each convolutional branch (DW and PW) is independently reparameterized during training and sequentially merged into a single-branch structure for inference. Inspired by these reparameterization methods, we propose a multiconvolution reparameterization strategy that efficiently merges hierarchical convolutions for faster inference while enhancing feature representation during the inference stage.

III. Data

In this study, we employ satellite BT data and rain gauge data provided by the China Meteorological Administration (CMA) to train and test the proposed RepNPE-Net. Fig. 1 presents the study domains, including the area around Shanghai (SH) and a large portion of Shandong Province (SD). SH spans a latitude range from 26.5° N to 32.5° N and a longitude range from 116° E to 122° E, while SD covers latitudes from 33.5° N to 39.5° N and longitudes from 114° E to 120° E. Among the regions, SH is equipped with dense rain gauge stations that provide long-term continuous observations, enabling the acquisition of reliable precipitation data. Therefore, in our experiments, data from SH are utilized to train the proposed network, while data from SD are used to evaluate its generalization and practical application ability. In addition, radar reflectivity data, including raw composite reflectivity (RCR) and composite reflectivity (CREF, the maximum base reflectivity factor), sourced from CMA's China New Generation Weather Radar Network (CINRAD), are adopted to visually compare NPE removal performance of various networks.

A. Multisource Data From Satellite, Radar, and Rain Gauges

BT data are derived from the FY-4A geostationary meteorological satellite, China's inaugural new-generation meteorological satellite, equipped with several advanced optical instruments, including the advanced geosynchronous radiation imager (AGRI), the Geosynchronous Interferometric Infrared Sounder, the Lightning Mapping Imager, and the Space Environment Package. The nadir point of the satellite is positioned at 104.7° E, providing extensive observation coverage that encompasses regions such as East Asia, Australia, and Indian Ocean. The AGRI boasts 14 spectral bands, offering a high spatial resolution ranging from 0.5 km for visible channels to 4 km for infrared channels. These bands cover a wide range of wavelengths, including visible channels from 0.47 to 0.65 μm, infrared channels from 3.725 to 13.5 μm, and near-infrared channels from 0.825 to 2.225 μm. Additionally, the

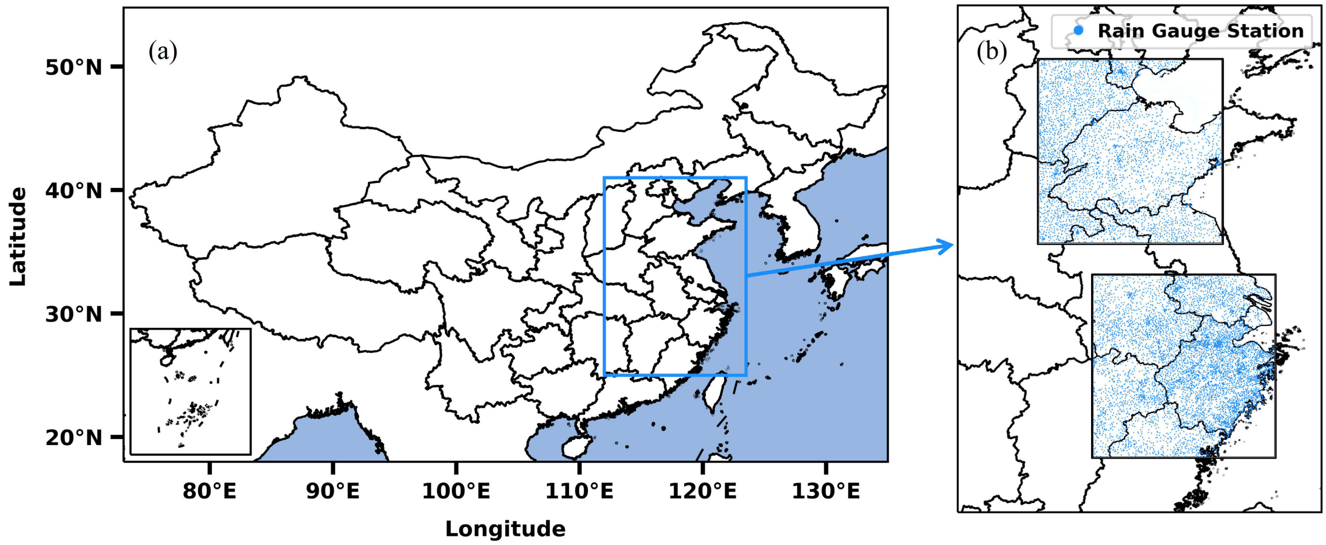

Fig. 1. Demonstration of study domains. (a) Map of China with the blue box highlighting Eastern China. (b) Distribution of rain gauge stations within study areas including SD (top rectangle) and SH (bottom rectangle).

satellite performs regular observations through two scanning operations: a comprehensive 15-min full-disk scan and a focused 5-min scan over the Chinese continental region. In this study, we utilize multichannel data from Band 08 (central wavelength of 3.75 μm), Band 10 (central wavelength of 6.90 μm), and Band 12 (central wavelength of 10.80 μm) in Chinese region for identifying and removing NPEs. These data are with a temporal and spatial resolution of 5 min and 4 km.

The rain gauge data are sourced from a comprehensive network of automatic weather stations (AWS) operated by the CMA. These stations are distributed across the entire Chinese region and record cumulative precipitation over specified intervals, such as 1, 3, and 24 h, with measurements logged at the beginning of each hour. In our experiments, we use 1-h cumulative rainfall data over SH and SD domains to train and test the proposed RepNPE-Net, further verifying its effectiveness and generalization ability.

RCR data include both PEs and NPEs, serving as the unprocessed input for NPE removal tasks. In contrast, CREF data are preprocessed using radar-based removal algorithms developed by CMA to eliminate NPEs. Due to their reliance on inherent radar parameters, these data still contain residual NPEs. CINRAD is a nationwide system consisting of over 123 S-band and 94 C-band Doppler weather radars, strategically distributed across various regions of China, including the densely populated eastern and coastal areas as well as the more remote northwestern and northeastern regions. The radar data utilized in this study have a temporal resolution of 6 min and a spatial resolution of 1 km.

B. Dataset Establishment

In this study, multichannel BT data and rain gauge data from SH and SD region are adopted. The data of SH area are collected during warm seasons from May to September 2022 to train and test the proposed RepNPE-Net, while those of the SD region are gathered from May 1 to 30, 2022 to test its generalization ability. We apply 4× bilinear interpolation [27] to align the BT data's spatial resolution with the radar data of 1 km for better comparison and analysis. The rain gauge data provide point-based measurements of cumulative rainfall over the previous hour. To align with this, we use BT data covering a 45×45 km area centered around each rain gauge location as input data. To ensure temporal consistency between satellite and rain gauge observations, the BT data closest in time to the rain gauge measurements are selected for dataset construction. A threshold of 0.2 mm/h is applied to the rain gauge data: values exceeding 0.2 mm/h are labeled as 0 (indicating rainfall points), while those below this threshold are labeled as 1 (indicating nonrainfall points). To ensure balanced training, precipitation and nonprecipitation points are selected in a 1:1 ratio to build the dataset.

Finally, in the SH area, we obtain 101 996 samples collected from May to September 2022, where 81 597 samples from May to August were used for training and validation (80% of the data) and 20 339 samples collected in September for testing (20% of the data). Then, we normalize these samples for the processes. In addition, 7866 samples collected from the SD region with the period of September 1–30 are adopted to test the generalization and practical application ability of the proposed RepNPE-Net. We employ satellite multichannel observations, rain gauge, and RCR data from Henan strong convective event on July 20, 2021 and nationwide event in China on July 26, 2024 to visually compare the performance of various networks and further demonstrate the effectiveness of the proposed method.

IV. Methodology

Fig. 2 illustrates the overall workflow of the proposed network. Especially, RepNPE-Net is designed as a reparameterization-based network that integrates dual-stream convolution modules (i.e., RepDCM and RepADCM) with HCR strategy, aiming to achieve accurate NPE recognition

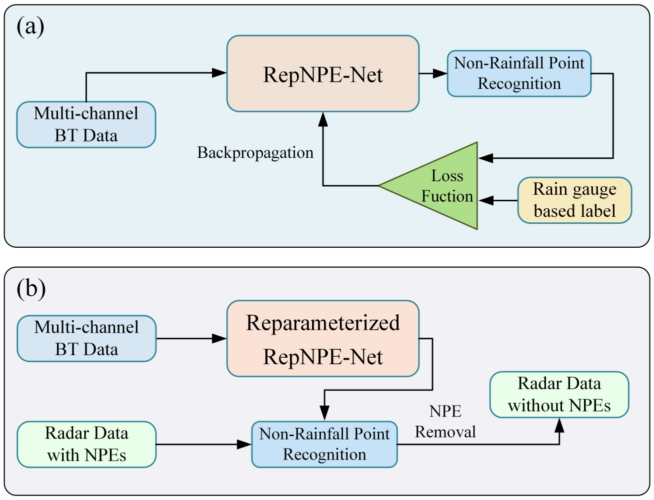

Fig. 2. Flowchart of training and inference stages for the proposed method. (a) Training phase. (b) Inference phase.

during training and efficient deployment during inference. The detailed network architecture and reparameterization strategy are introduced in Sections IV-A and IV-B

A. Architecture of RepNPE-Net

Fig. 3 illustrates the architecture of the proposed RepNPE-Net, along with details of specific sub-modules and sub-blocks. The network is mainly composed of initial convolution module, three downsample modules (i.e., Downsample-1, Downsample-2, and Downsample-3), four stages based on RepDCM and RepADCM modules, and two dense layers, including Dense-128 and Dense-1, which are further described as follows.

1) Initial Convolution Module: The module consists of a standard 3×3 convolution, a 3×3 DW convolution, and ReLU activation functions [28]. The number of convolutional kernels in the layers is both set to 32. The stride is set to 1 and 2.

2) Stage: In the proposed RepNPE-Net, various stages (Stages 1–4) are designed to capture complex feature representations of satellite data. In each stage, we introduce two reparameterized modules based on dual-stream convolution structure, including RepDCM and RepADCM, whose numbers are assigned to 1, 1, 7, and 1 in Stages 1–4, respectively, as shown in Fig. 3. The larger number of modules in Stage 3 is intentionally designed to enhance the network's capacity for extracting complex high-level semantic features between satellite observations and NPEs. Each module adopts a dual-stream convolution structure that integrates a standard residual convolutional block with DS residual convolutional block, aiming to balance computational efficiency and feature extraction capability. Specifically, the standard convolution is employed to capture complex and high-dimensional feature interactions, providing robust representation capabilities. In contrast, the DS convolution decouples spatial and channel-wise convolutions, focusing on efficient local feature extraction and significantly reducing computational costs. By combining these two components, the modules achieve an optimal balance between performance and efficiency, enhancing the representation of nonprecipitation characteristics while reducing inference time. The number of convolutional kernels in the structure in RepADCM and RepDCM modules is set to 32, 64, 128, and 256 in Stages 1–4, respectively. The stride is all assigned to 1. Notably, during the inference stage, various reparameterization techniques are employed to reduce computational complexity and memory requirements while maintaining the model's performance. More details can be found in Section IV-B.

After the dual-stream convolution structure, inspired by [29], we design a PELA block in RepADCM modules to preserve important spatial position information. Unlike fixed positional encoding, PELA incorporates a learnable positional embedding (LPE) optimized during the training process, enabling the model to adaptively capture spatial relationships and improve feature representation. As illustrated in Fig. 4, the positional embedding is directly added to the input feature map $\mathbf{X}$, which can be defined as follows:

$$\mathbf{X}' = \mathbf{X} + \mathbf{P} \tag{1}$$

where $\mathbf{P} \in \mathbb{R}^{C \times H \times W}$ (where C, H, and W denote the channel, height, and width of the feature map, respectively) represents the LPE. $\mathbf{X}'$ is the output feature map after adding the position embedding. The embedded features are then processed through two parallel branches along the X and Y dimensions. The first branch applies X-axis average pooling followed by a 1×1 convolution, group normalization, and a sigmoid activation. Similarly, the second branch utilizes Y-axis average pooling with the same subsequent operations, including 1×1 convolution, group normalization, and sigmoid activation. The group number in the branches is both set to 16. The number of convolutional kernels in the convolutions in the stages is set to 32, 64, 128, and 256. The outputs from both branches are added to the input features to obtain the output features of PELA block. Finally, a feed-forward network (FFN) [30] is adopted to obtain the output of RepADCM block. The component is composed of two 1×1 convolution operations, a ReLU activation function, and a residual connection. The kernel number of the convolutions in the stages is set to [64, 128, 256, 512] and [32, 64, 128, 256]. The stride is all assigned to 1.

Similarly, for the RepDCM modules, after the dual-stream convolution structure, an FFN network is directly adopted to obtain the output. The parameter settings are the same as those of RepADCM.

3) Downsample Module: Between four stages, three downsample modules, including Downsample-1, Downsample-2, and Downsample-3, are designed to progressively reduce the spatial dimensions while preserving essential feature information. Each block consists of a RepDCM module, a 3×3 DW convolution, a 1×1 convolution, and an FNN network. The RepDCM's parameter settings in three modules are the same as those in Stages 1–3. For the DW and 1×1 convolutions, the number of kernels is configured as 64 128 and 256 across the three modules, with strides set to 1 and 2, respectively. In the FNN, the first 1×1 convolution employs 128 256 and 512 kernels in the three modules, while the second convolution

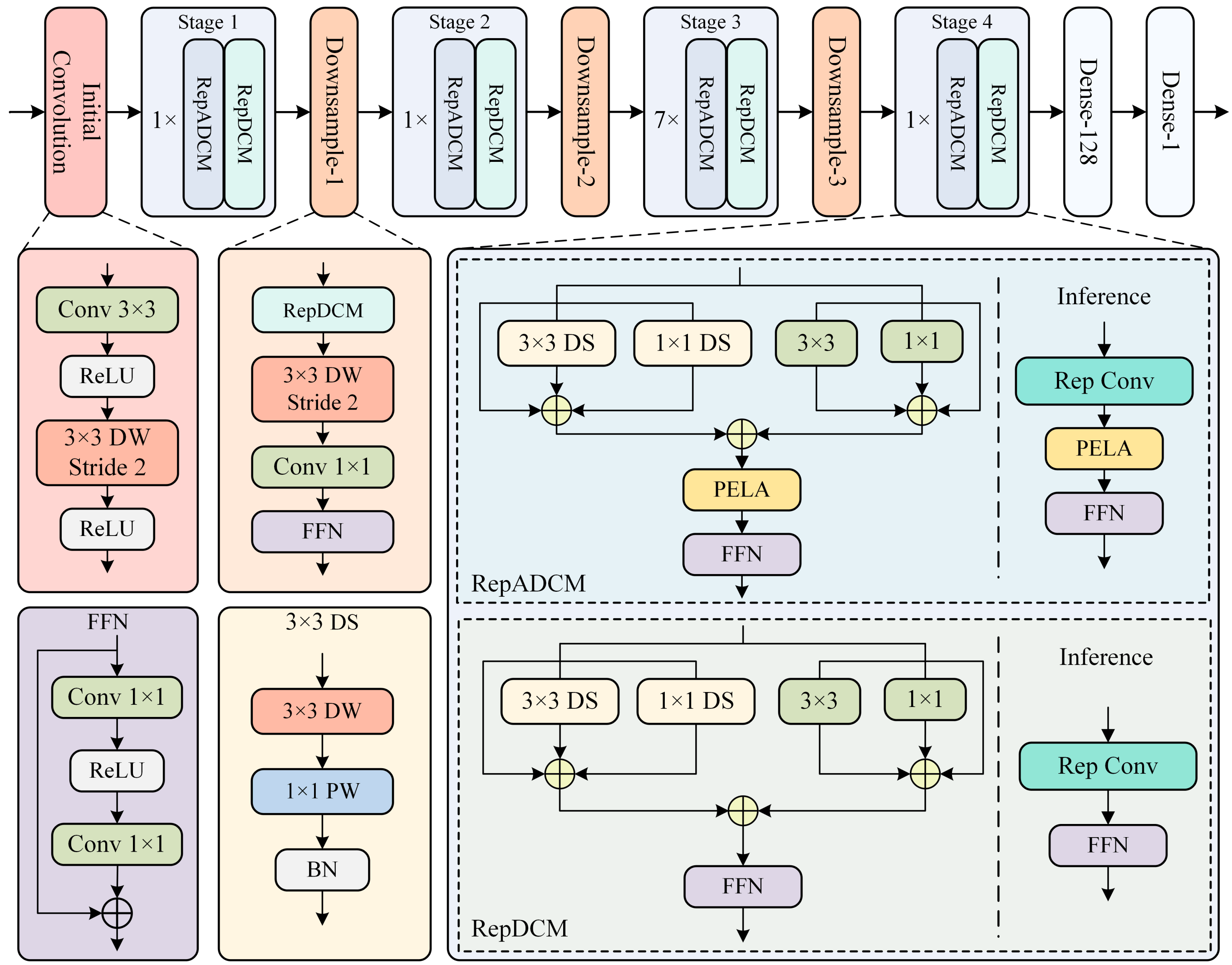

Fig. 3. Structure of the proposed RepNPE-Net.

uses 64 128 and 256 kernels. The stride for both convolutions is set to 1.

After Stage 4, two dense layers, including Dense-128 and Dense-1, are used to obtain raw scores (logits) of NPE recognition results. Then, a threshold of 0.5 is applied to perform binary classification, where predictions greater than or equal to 0.5 are classified as nonrainfall points (label 1) and predictions less than 0.5 are seen as rainfall points (label 0).

B. Reparameterization During Inference Stage

During the inference stage, to reduce computational complexity, various reparameterization techniques are employed to simplify the dual-stream convolution structure in the RepDCM and RepADCM modules, as shown in Fig. 5. Conventional DS convolution reparameterization approaches typically reparameterize DW and PW convolution structures separately and then combine them sequentially into a single-branch convolution [13]. While this reduces computational burden to some extent, the structure still retains separate DW and PW components and does not fully explore the efficiency potential of sequential DW-PW convolution. To overcome this limitation, we propose a novel HCR strategy. Unlike conventional approaches that merely simplify multibranch structures, HCR uniquely fuses multibranch and multiconvolution operations (e.g., DW convolution and PW convolution) into a single equivalent convolution, significantly streamlining computation and enhancing inference efficiency. For the standard residual convolutional block within the dual-stream structure, multibranch reparameterization is applied to simplify its multibranch operations. Finally, the reparameterized outputs of DS and standard residual blocks are fused to produce the final equivalent convolution of the entire dual-stream structure.

1) HCR Reparameterization of DS Convolution Block: As shown in the green rectangle in Fig. 5(a), the block includes 3×3 DS convolution branch, 1×1 DS convolution branch, and residual branch. Each DS convolution operation consists of two steps, including DW convolution and 1×1 PW convolution. Let C_{in} denote the channel number of input feature map X, and the former operates on each input channel independently as follows:

$$Y_d = W_d * X + b_d \tag{2}$$

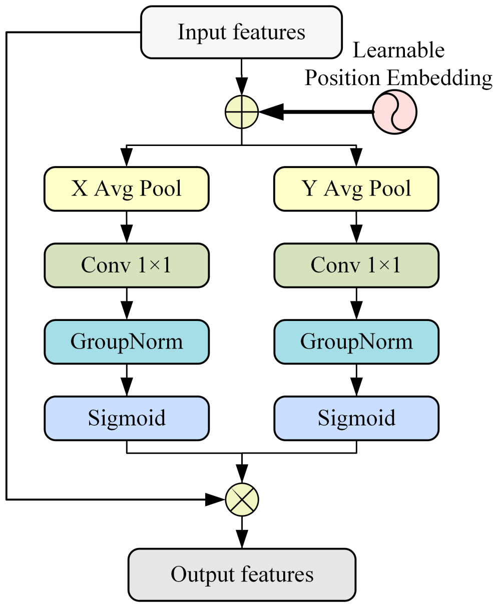

Fig. 4. Flowchart of position embedding-based efficient local attention (PELA).

where X represents input data. W_d is DW convolutional kernel with the size of $K \times K \times C_{\text{in}}$ and Y_d is the output of DS convolution, where $K \times K$ indicates the kernel size of DW convolution. Then, the PW convolution is adopted to mix information from different input channels as follows:

$$Y_{dp} = W_p * Y_d + b_p \tag{3}$$

where W_p is the PW convolution kernel with the size of $1 \times 1 \times C_{\text{in}} \times C_{\text{out}}$ and Y_{dp} is the output of PW convolution. The DS convolution operation can be merged from the steps as follows:

$$\begin{aligned} Y_{dp} &= W_p * (W_d * X + b_d) + b_p \\ &= \left(W_p * W_d\right) * X + \left(W_p * b_d + b_p\right) \\ &= W_{dp} * X + b_{dp} \end{aligned} \tag{4}$$

where $W_{dp} = W_p * W_d$ and $b_{dp} = W_p * b_d + b_p$ denotes the equivalent convolution kernel and bias of DS convolution operation. Through the above algebraic operations, the multiconvolution DS structure comprising DW and PW convolutions is transformed into a single equivalent convolution, significantly enhancing inference speed while preserving the network's representation ability.

Based on multiconvolution reparameterization, a 3×3 DS convolution branch can be reparameterized into an equivalent DS convolution expressed as $Y_{dp}^1 = W_{dp}^1 * X + b_{dp}^1$. For a 1×1 DS convolution branch, we can obtain the equivalent DS convolution (symbolized as $Y_{dp}^2 = W_{dp}^2 * X + b_{dp}^2$) by padding zeros around the 1×1 DW convolutional kernel W_O^2. Similarly, residual branch is seen as 1×1 convolution with the convolutional kernel of unit matrix, which is then adopted to obtain equivalent 3×3 convolution (represented as $Y_{dp}^3 = W_{dp}^3 * X + b_{dp}^3$, where $b_{dp}^3 = 0$) by padding zeros around the equivalent 1×1 convolutional kernel. Then, these equivalent convolutions are fed into the BN operation to obtain the fused result of each branch as follows:

$$\begin{aligned} Y_{ds,i}^f &= \gamma_i \times \frac{Y_{dp}^i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i \\ &= \gamma_i \times \frac{W_{dp}^i * X + b_{dp}^i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i \\ &= \left(\frac{\gamma_i \times W_{dp}^i}{\sqrt{\sigma_i^2 + \epsilon}}\right) * X + \left(\frac{\gamma_i \times \left(b_{dp}^i - \mu_i\right)}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i\right) \end{aligned} \tag{5}$$

where μ_i and σ_i are mean and variance values of Y_{dp}^i ($i \in 1, 2, 3$). γ and β represent learnable scale and shift parameter, which are adopted to adjust the normalized values and recenters the normalized values, respectively. ϵ is a small constant added for numerical stability, which is set to 10^{-5} in our experiments. Then, the fused convolution of each branch can be represented as follows:

$$Y_{ds,i}^f = W_{ds}^i * X + b_{ds}^i \tag{6}$$

where W_{ds}^i and b_{ds}^i are weights and biases of convolutional kernels in ith fused convolution, which are defined as follows:

$$W_{ds}^i = \frac{\gamma_i \times W_p^i * W_d^i}{\sqrt{\sigma_i^2 + \epsilon}} \tag{7}$$

$$b_{ds}^i = \frac{\gamma_i \times \left(b_{dp}^i - \mu_i\right)}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i. \tag{8}$$

Finally, we can obtain the reparameterized output of DS convolution from multibranch fused convolutions as follows:

$$\begin{aligned} Y_{ds} &= \sum_{i=1}^{3} \left(W_{ds}^i * X + b_{ds}^i\right) \\ &= W_{ds} * X + b_{ds} \end{aligned} \tag{9}$$

where $W_{ds} = W_{ds}^1 + W_{ds}^2 + W_{ds}^3$ and $b_{ds} = b_{ds}^1 + b_{ds}^2 + b_{ds}^3$ represent the weights and biases of fused convolutional kernels from all branches, respectively. To clearly illustrate the derivation of fused convolution weight and bias, we provide an algorithmic overview in Algorithm 1. The algorithm summarizes the process of merging DW, PW, and residual branches into a single equivalent convolution during inference, offering a concise understanding of the proposed HCR reparameterization workflow.

2) Multibranch Reparameterization of Standard Convolution Block: Similar to the DS convolution block, the standard convolution block consists of 3×3 convolution branch, 1×1 convolution branch, and residual branch, as shown in the orange rectangle in Fig. 5(b). Inspired by [6], we adopt the same reparameterization strategy to transform a multibranch structure into a single-branch architecture. The standard convolution operation can be defined as follows:

$$Y = W * X + b \tag{10}$$

where W and b denote the weight matrix and bias, respectively. For the standard 3×3 convolution branch, we use its original

Algorithm 1 HCR for DS Convolution Block

Require: 3×3 DS branch weights $(W_d^1, W_p^1, b_d^1, b_p^1)$, original 1×1 DS branch weights $(W_o^2, W_d^2, b_d^2, b_p^2)$, original residual branch, BN parameters $(\gamma, \beta, \mu, \sigma)$, constant ϵ

Ensure: HCR weight W_{ds} and bias b_{ds}

for $i = 1$ to 3 **do**
  if $i = 1$ **then**
    Compute equivalent weight: $W_{dp}^1 \leftarrow W_p^1 * W_d^1$
    Compute equivalent bias: $b_{dp}^1 \leftarrow W_p^1 * b_d^1 + b_p^1$
  else if $i = 2$ **then**
    $W_p^2 \leftarrow$ Pad 1×1 DW convolutional kernel W_o^2
    Compute equivalent weight: $W_{dp}^2 \leftarrow W_p^2 * W_d^2$
    Compute equivalent bias: $b_{dp}^2 \leftarrow W_p^2 * b_d^2 + b_p^2$
  else
    Obtain equivalent weight: $W_{dp}^3 \leftarrow$ Pad unit matrix
    Obtain equivalent bias: $b_{dp}^3 \leftarrow 0$
  end if
  Branch fusion weight: $W_{ds}^i \leftarrow \frac{\gamma_i \times W_{dp}^i}{\sqrt{\sigma_i^2 + \epsilon}}$
  Branch fusion bias: $b_{ds}^i \leftarrow \frac{\gamma_i \times (b_{dp}^i - \mu_i)}{\sqrt{\sigma_i^2 + \epsilon}} + \beta_i$
end for
Compute HCR weight: $W_{ds} = W_{ds}^1 + W_{ds}^2 + W_{ds}^3$
Compute HCR bias: $b_{ds} = b_{ds}^1 + b_{ds}^2 + b_{ds}^3$
return W_{ds}, b_{ds}

convolutional kernel for reparameterization, which is denoted as $Y_1 = W_1 * X + b_1$. For 1×1 convolution branch, we can obtain the equivalent 3×3 convolution (symbolized as $Y_2 = W_2 * X + b_2$) by padding zeros around the 1×1 convolutional kernel. Similarly, the residual branch is seen as 1×1 convolution with the convolutional kernel of unit matrix, which is then adopted to obtain equivalent 3×3 convolution (represented as $Y_3 = W_3 * X + b_3$) by padding zeros around the equivalent 1×1 convolutional kernel. Then, these equivalent convolutions are fed into the BN operation to obtain the fused result of each branch as follows:

$$\begin{aligned} Y_i^f &= \gamma_i' \times \frac{Y_i - \mu_i}{\sqrt{\sigma_i'^2 + \epsilon}} + \beta_i' \\ &= \gamma_i' \times \frac{W_i * X + b_i - \mu_i}{\sqrt{\sigma_i'^2 + \epsilon}} + \beta_i' \\ &= \left(\frac{\gamma_i' \times W_i}{\sqrt{\sigma_i'^2 + \epsilon}} \right) * X + \left(\frac{\gamma_i' \times (b_i - \mu_i')}{\sqrt{\sigma_i'^2 + \epsilon}} + \beta_i' \right) \end{aligned} \tag{11}$$

where μ_i' and σ_i' are mean and variance values of Y_i ($i \in \{1, 2, 3\}$). γ' and β' represent learnable scale and shift parameter. Based on the standard convolution operation, the fused convolution of each branch can be represented as follows:

$$Y_i^f = W_i^f * X + b_i^f \tag{12}$$

where W_i^f and b_i^f are weights and biases of convolutional kernels in the ith fused convolution, which is defined as follows:

$$W_i^f = \frac{\gamma_i' \times W_i}{\sqrt{\sigma_i'^2 + \epsilon}} \tag{13}$$

$$b_i^f = \frac{\gamma_i' \times (B_i - \mu_i')}{\sqrt{\sigma_i'^2 + \epsilon}} + \beta_i'. \tag{14}$$

Finally, we can obtain the reparameterized output of standard convolution block from multibranch fused convolutions as follows:

$$\begin{aligned} Y_s &= \sum_{i=1}^{3} W_i^f * X + b_i^f \\ &= W_s * X + b_s \end{aligned} \tag{15}$$

where $W_s = W_1^f + W_2^f + W_3^f$ and $b_s = b_1^f + b_2^f + b_3^f$ represent the weights and bias of fused convolutional kernels from all branches.

3) Final Fusion: As illustrated in Fig. 5(c), based on equivalent convolutional kernels and biases of standard and DS convolution operations, we can further obtain the single convolution output of dual-stream convolution structure in the RepDCM and RepADCM modules as follows:

$$\begin{aligned} Y_{rep} &= Y_s + Y_{ds} \\ &= W_s * X + b_s + W_{ds} * X + b_{ds} \\ &= (W_s + W_{ds}) * X + (b_s + b_{ds}) \\ &= W_{rep} * X + b_{rep} \end{aligned} \tag{16}$$

where $W_{rep} = W_s + W_{ds}$ and $b_{rep} = b_s + b_{ds}$ are the reparameterized convolutional kernel and bias of the dual-stream convolution structure, respectively.

As shown in Fig. 3, through the usage of reparameterization technologies, multiconvolution and multibranch structures of RepDCM and RepADCM modules are converted into single-branch hierarchical structures, significantly enhancing inference speed while preserving performance.

V. Experiments and Analysis

A. Training Details

During the training stage, multichannel BT data from the FY-4A satellite and rain gauge-based labels over SH region are employed to train, validate, and test the proposed network for accurate NPE recognition. In addition, the BCE loss function is adopted to train RepNPE-Net as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{j=1}^{N} \left[y_j \log \left(\sigma \left(z_j \right) \right) + \left(1 - y_j \right) \log \left(1 - \sigma \left(z_j \right) \right) \right] \tag{17}$$

where N is the total number of samples, y_j ($y \in \{0, 1\}$) represents the ground truth label for sample j, and z_j is the logit output of the model. $\sigma(z) = 1/(1 + e^{-z})$ denotes logistic sigmoid function. In addition, we use Adam optimization [31] with initial learning rate 1×10^{-4} to optimize the regression object. The model is trained for 200 epochs with a batch size of 128 to obtain final estimates of nonrainfall points.

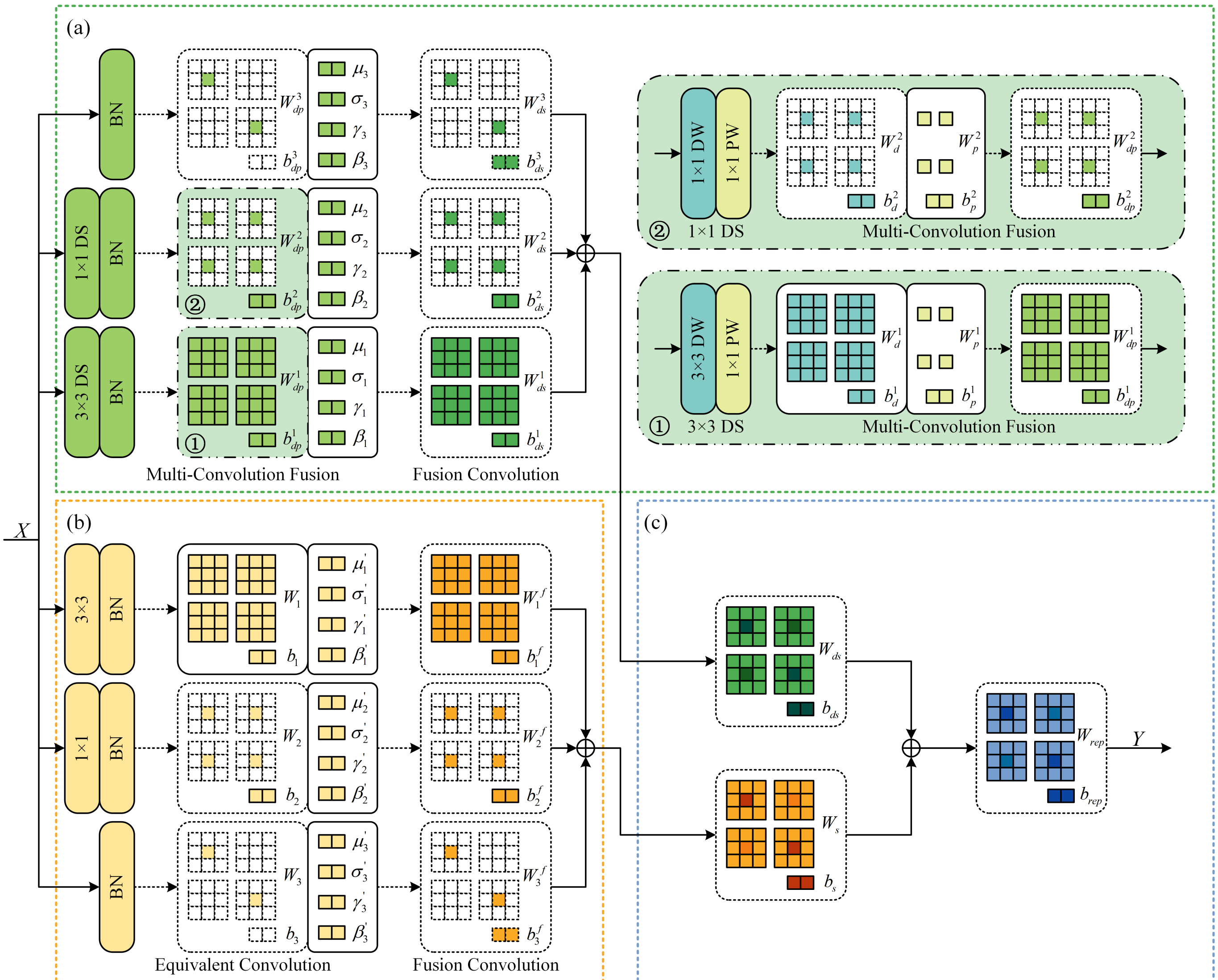

Fig. 5. Schematic of structural reparameterization in inference stage. In this case, the numbers of input channels and convolutional kernels in DS and standard convolutions are all set to 2. (a) HCR. (b) Multibranch reparameterization. (c) Final fusion.

B. Comparison Methods

Eight experiments, U-Net, VGG, ResNet, RepVGG, RepViT, UniRepLKNet, InternImage, and RepNPE-Net, are adopted to compare the performance of various networks as follows.

1) *U-Net [23]:* A widely used CNN network with an encoder–decoder structure and skip connections, enabling efficient information preservation and detailed feature recovery during propagation;
2) *VGG [32]:* A classic CNN architecture comprising 16 layers designed to learn spatial hierarchies through a series of small 3×3 convolution kernels.
3) *ResNet [33]:* An 18-layer CNN architecture with residual blocks designed to maintain pristine information and stable training through residual connections;
4) *RepVGG [6]:* A CNN architecture utilizing a simplified structure with efficient residual blocks designed to streamline feature extraction and maintain stable training, offering a balance between computational efficiency and performance.
5) *RepViT [7]:* The baseline network in our study is a reparameterized CNN network that transforms multibranch structures into an efficient single-path during inference, optimizing speed and performance.
6) *UniRepLKNet [10]:* A reparameterized large-kernel CNN architecture unifies diverse convolutional branches into single large-kernel convolution, enabling efficient feature extraction across various scales.
7) *InternImage [34]:* A versatile vision architecture employing deformable convolution-based blocks designed to capture adaptive features and achieve strong performance across diverse vision tasks;
8) *RepNPE-Net:* The proposed network in this study incorporates dual-stream convolution structure-based RepDCM and RepADCM modules for better feature representation, which are then optimized by HCR reparameterization technology to reduce computational

TABLE I
DEFINITION OF VARIOUS INDEX PARAMETERS

		Observation	
		T	F
Prediction	T	TP	FP
	F	FN	TN

complexity while maintaining performance during inference.

During the training stage, we modified U-Net, VGG, ResNet, RepVGG, and RepViT by adding Dense layers after the original network architecture to adjust for NPE removal tasks and retrained them on our dataset to obtain experimental results presented in this study. Notably, the loss function, optimization algorithm, and other hyperparameters are consistent with RepNPE-Net in this phase.

C. Indices

In this study, we evaluate the classification performance using eight assessment metrics derived from the contingency table approach, which are widely utilized in both atmospheric and computer vision fields. Specifically, accuracy (ACC), precision (PREC), recall, and $F1$-score ($F1$) are commonly employed in computer vision, assessing the model's precision and recall capabilities. In contrast, probability of detection (POD), false alarm rate (FAR), critical success index (CSI) [35], and Heidke skill score (HSS) [36] are widely adopted in atmospheric sciences. High values of ACC, PREC, Recall, F1, POD, CSI, and HSS, alongside low FAR, indicate strong model performance. These metrics are defined as follows:

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}} \tag{18}$$

$$\mathrm{PREC} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{19}$$

$$\mathrm{RECALL} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{20}$$

$$\mathrm{F1} = \frac{2\cdot\mathrm{TP}}{2\cdot\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{21}$$

$$\mathrm{POD} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{22}$$

$$\mathrm{FAR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TP}} \tag{23}$$

$$\mathrm{CSI} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{24}$$

$$\mathrm{HSS} = \frac{2\cdot(\mathrm{TP}\cdot\mathrm{TN}-\mathrm{FN}\cdot\mathrm{FP})}{(\mathrm{FN}+\mathrm{TP})\cdot(\mathrm{TN}+\mathrm{FN})+(\mathrm{TP}+\mathrm{FP})\cdot(\mathrm{TN}+\mathrm{FP})} \tag{25}$$

where TP, TN, FN, and FP denote the true positive, true negative, false negative, and false positive, respectively. The definition of these parameters can be found in Table I, where "T" and "F" represent true and false classification results in rainfall point estimation missions.

D. Results and Analysis

Table II presents overall performance comparison results of different networks over testing dataset across multiple evaluation metrics, with best indices highlighted in bold. U-Net, with its simplistic network structure, struggles to extract the complex spatial relationships required for the NPE removal task. Thus, it consistently shows lower performance than other networks across all assessment indices, exhibiting the lowest ACC, PREC, F1, RECALL, FAR, CSS, and HSS values, along with the highest FAR values. Due to a deeper and more uniform convolutional structure, VGG achieves better performance than U-Net but still lacks the ability to retain critical information during network propagation. ResNet addresses these issues by introducing residual connections and thus achieves better performance than U-Net and VGG in most assessment indices. RepVGG further improves on ResNet by adopting a simplified yet efficient architecture, which enhances its ability to extract and preserve key features, leading to better ACC, PREC, F1, FAR, and HSS indices. Building upon RepVGG, RepViT introduces a more optimized hierarchical convolutional structure, which further refines feature extraction and task adaptation, resulting in superior performance across all evaluation indices. Based on this approach, UniRepLKNet incorporates reparameterized large-kernel convolutions to enlarge the receptive field and strengthen global context modeling, achieving comparatively better indices than RepVGG and RepViT. InternImage further enhances feature extraction ability by employing deformable convolution-based blocks, allowing it to capture diverse spatial patterns and yielding improved performance across multiple indices. However, without PELA attention block and dual-stream convolution structure, these methods still fall short of the proposed RepNPE-Net. By introducing the RepDCM and RepADCM modules integrating the components, RepNPE-Net achieves the best performance among all comparison methods. Specifically, it outperforms the baseline network (RepViT) with significant improvements in ACC (+0.022), RECALL (+0.058), POD (+0.058), CSI (+0.045), and HSS (+0.046), demonstrating the superiority of the proposed method for NPE removal.

For qualitative comparison, Fig. 6 provides a visualized NPE removal case using U-Net, ResNet, RepViT, and proposed RepNPE-Net, along with corresponding satellite BT observation with central wavelength of 10.80 μm, rain gauge observation, RCR, and CREF data. It is evident that CREF data and comparison networks can effectively remove NPEs with reflectivity values below approximately 30 dBZ. For other methods, the removal ability differs significantly. Notably, as shown in the red rectangles in Fig. 6(c), a large convective storm can be observed. However, rain gauge observation in Fig. 6(b) indicates that the hourly cumulative precipitation in the area is close to zero, identifying it as a nonprecipitation region. Compared to other networks, including U-Net, ResNet, and RepViT, the proposed RepNPE-Net demonstrates superior performance by completely removing NPEs within the rectangle, highlighting the effectiveness of the proposed RepDCM and RepADCM modules.

To enhance the interpretability of RepNPE-Net, Fig. 7 provides visualized feature response maps of Stage 1 with and without PELA attention. As shown in the figure, the attention maps assign higher weights to physically relevant

TABLE II
OVERALL PERFORMANCE COMPARISON OF DIFFERENT NETWORKS OVER TESTING DATASET

	ACC	PREC	RECALL	F1	POD	FAR	CSI	HSS
U-Net	0.706	0.709	0.479	0.628	0.479	0.291	0.457	0.431
VGG	0.710	0.728	0.609	0.697	0.609	0.272	0.535	0.419
ResNet	0.703	0.739	0.628	0.679	0.628	0.261	0.514	0.406
RepVGG	0.717	0.818	0.559	0.684	0.559	0.182	0.497	0.434
RepViT	0.725	0.824	0.571	0.699	0.571	**0.176**	0.509	0.449
UniRepLKNet	0.726	0.814	0.587	0.682	0.587	0.187	0.518	0.453
InternImage	0.736	0.792	**0.629**	0.708	**0.629**	0.208	0.547	0.471
RepNPE-Net	**0.747**	**0.825**	**0.629**	**0.713**	**0.629**	**0.176**	**0.554**	**0.495**

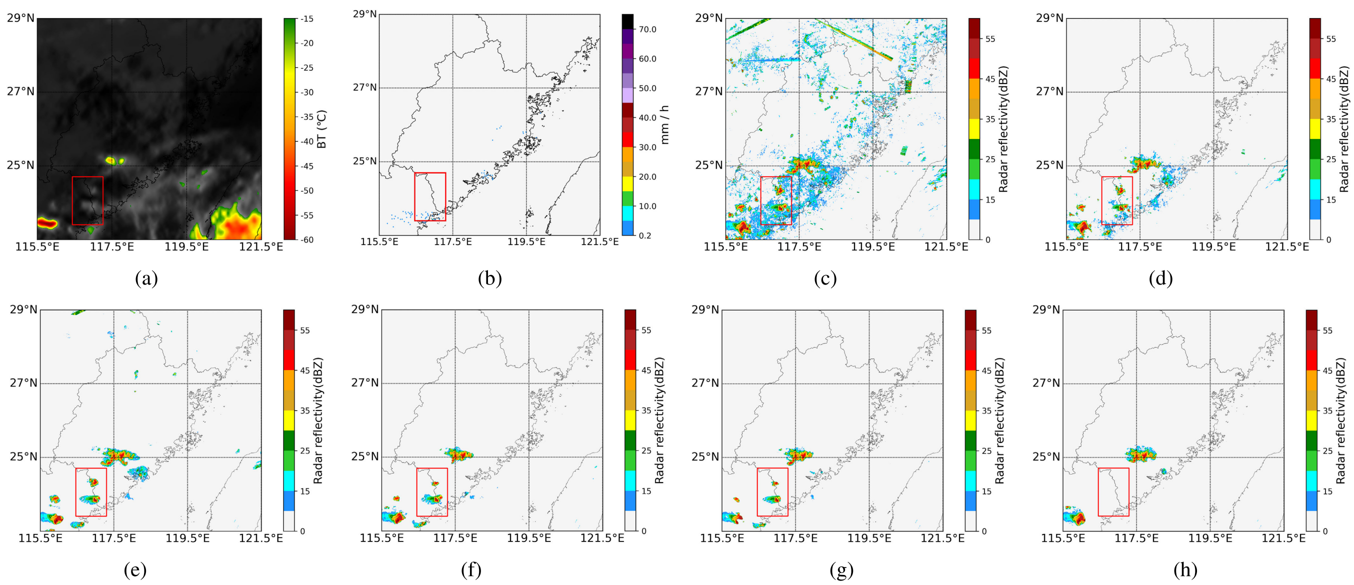

Fig. 6. NPE removal case using various networks at 13:10 UTC on September 1, 2023. (a) Satellite observation (10.80 μm). (b) Rain gauge observation. (c) RCR. (d) CREF. (e) U-Net. (f) ResNet. (g) RepViT. (h) RepNPE-Net.

TABLE III
PERFORMANCE COMPARISON OF ABLATION EXPERIMENTS WITH DIFFERENT SCHEMES

	ACC	PREC	RECALL	F1	POD	FAR	CSI	HSS
Scheme-1	0.725	0.824	0.571	0.699	0.571	**0.176**	0.509	0.449
Scheme-2	0.739	0.822	0.619	0.707	0.619	0.258	0.523	0.417
Scheme-3	0.742	**0.827**	0.625	0.711	0.625	0.193	0.551	0.483
Scheme-4	**0.747**	0.825	**0.629**	**0.713**	**0.629**	**0.176**	**0.554**	**0.495**

nonprecipitation regions rather than convective cloud tops with low BT values or cloud boundaries with strong temperature gradients, indicating that the model tends to focus on meteorological features for NPE identification rather than random statistical correlations. In addition, the dual-stream RepDCM and RepADCM modules are designed to extract complementary spatial features. The RepDCM module focuses on local spatial patterns of NPEs, allowing the network to capture subtle variations in small-scale nonprecipitation structures. In contrast, the attention-based RepADCM module emphasizes globally distributed nonprecipitation regions, integrating large-scale contextual information to improve overall recognition accuracy.

E. Ablation Experiments

In ablation experiments, four schemes are designed to verify the performance of the proposed RepNPE-Net as follows: 1) *scheme-1:* the baseline network utilized in this study; 2) *scheme-2:* the baseline network enhanced with the dual-stream convolution structure; 3) *scheme-3:* the network equipped with both the dual-stream convolution structure and the ELA attention block; and 4) *scheme-4:* the proposed network integrating PELA-based RepDCM and RepADCM modules. The experimental results are presented in Table III. It can be found that the progressive improvement can be obtained in most indices as the experimental items are successively incorporated into the baseline network, reflecting the effectiveness of dual-stream convolution structure and the proposed PELA block.

In addition, we specifically varied parameters such as various attention blocks, including SE, CBAM, ELA, and PELA blocks, and different embedding methods, including sinusoidal positional encoding (SPE), relative positional encoding (RPE), and LPE to evaluate the performance of the proposed

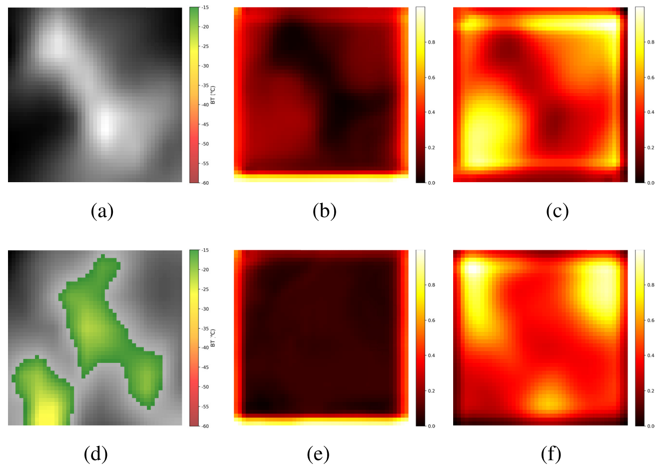

Fig. 7. Visualized feature response maps of Stage 1. (a) Satellite image. (b) Without PELA. (c) With PELA. (d) Satellite image. (e) Without PELA. (f) With PELA.

TABLE IV
PERFORMANCE COMPARISON WITH VARIOUS ATTENTION BLOCKS

Attention block	POD	FAR	CSI	HSS
SE	0.624	0.184	0.547	0.483
CBAM	0.620	0.171	0.550	0.492
ELA	0.625	0.193	0.551	0.483
PELA	**0.629**	**0.176**	**0.554**	**0.495**

TABLE V
PERFORMANCE COMPARISON WITH DIFFERENT POSITION EMBEDDING METHODS

Embedding method	POD	FAR	CSI	HSS
SPE	0.623	0.200	0.538	0.466
RPE	0.627	0.187	0.549	0.484
LPE	**0.629**	**0.176**	**0.554**	**0.495**

network, without other parameters and components changed. The experimental results are listed in Tables IV and V. In the tables, we apply four indices, including POD, FAR, CSI, and HSS, for comparison. It is obvious that the network with PELA block and LPE achieves the best performance across all comparison methods, which are adopted to build the proposed RepNPE-Net.

F. Case Study of Henan Torrential Rainstorm

Extreme rainfall events, characterized by high intensity and rapid onset, represent a significant challenge for meteorological monitoring and prediction. However, the presence of NPEs in radar observations further exacerbates this difficulty, leading to inaccuracies such as overestimation or misclassification of precipitation intensity and extent. Such misrepresentation poses significant challenges to both forecasting precision and disaster response effectiveness. Fig. 8 shows a visualized NPE removal case of Henan torrential rainstorm at 03:00 UTC on July 20, 2021. The figure provides NPE removal results using U-Net, ResNet, RepViT, and the proposed RepNPE-Net, alongside corresponding satellite observation, rain gauge observation, RCR, and CREF data. At that time, the event led to heavy rainfall across vast regions of North China, resulting in significant economic losses and casualties. Compared with satellite, rain gauge, and RCR data, it is evident that all comparison methods can effectively remove NPEs, offering comparatively higher quality radar data than CREF data. Notably, as shown in the red ellipse in Fig. 8(c), a large range of NPEs is visible, even though there is no rainfall in the region [see Fig. 8(b)]. Among the visualized results, the proposed RepNPE-Net, benefiting from RepDCM and RepADCM modules based on dual-stream convolution structure, successfully eliminates all NPEs in the ellipse area, further underscoring its superiority. This case underscores the critical role of the proposed network in recognizing and removing NPEs from radar RCR data, particularly in regions with complex clutter patterns and sparse rain gauge coverage. Such advancements are essential for improving the accuracy of precipitation estimation, supporting severe weather forecasting, and enhancing disaster response efforts during extreme meteorological events.

G. Generalization Performance Tests and Application

To test the generalization performance of the proposed RepNPE-Net, the trained network from the SH area is directly applied to identify and remove NPEs over the SD area. Mean assessment values, including ACC, PRED, RECALL, F1, POD, FAR, CSI, and HSS, are adopted for quantitative comparison of various networks. The experimental results are presented in Table VI. Evidently, except for FAR and PREC, the proposed network outperforms all comparison models across other evaluation metrics, indicating superior generalization performance. Notably, compared to the baseline network (RepViT), RepNPE-Net achieves significant improvements in ACC (0.654 versus 0.684), PREC (0.730 versus 0.778), RECALL (0.476 versus 0.506), and F1 (0.576 versus 0.613), demonstrating the effectiveness of the RepDCM and RepADCM modules with dual-stream convolution and PELA blocks.

Fig. 9 illustrates a visualized NPE removal case of China region using RepViT and the proposed RepNPE-Net at 03:42 UTC on July 26, 2024. The figure also includes corresponding satellite observation, rain gauge, RCR, and CREF data. It is evident that compared to satellite and rain gauge observations, both RepViT and the proposed network can remove more NPEs than CREF data in the nationwide region. Furthermore, as highlighted in the red rectangles, RepNPE-Net outperforms RepViT by eliminating more NPEs, underscoring the effectiveness of the proposed network in enhancing radar data quality.

H. Computational Complexity

In this study, all experiments are conducted on a Linux server equipped with two Intel Xeon E5-2680 CPUs and four NVIDIA GeForce RTX 3090 GPUs, each with 24-GB memory. The software implementation is based on TensorFlow 2.80 with Keras, running on CUDA 11.8 and cuDNN 8.5 under Ubuntu 22.04. To verify the computational efficiency of the proposed RepNPE-Net and comparison networks, the parameter number of each model and mean inference times over testing dataset and the entire China region are adopted. The experimental results are listed in Table VII. In the table,

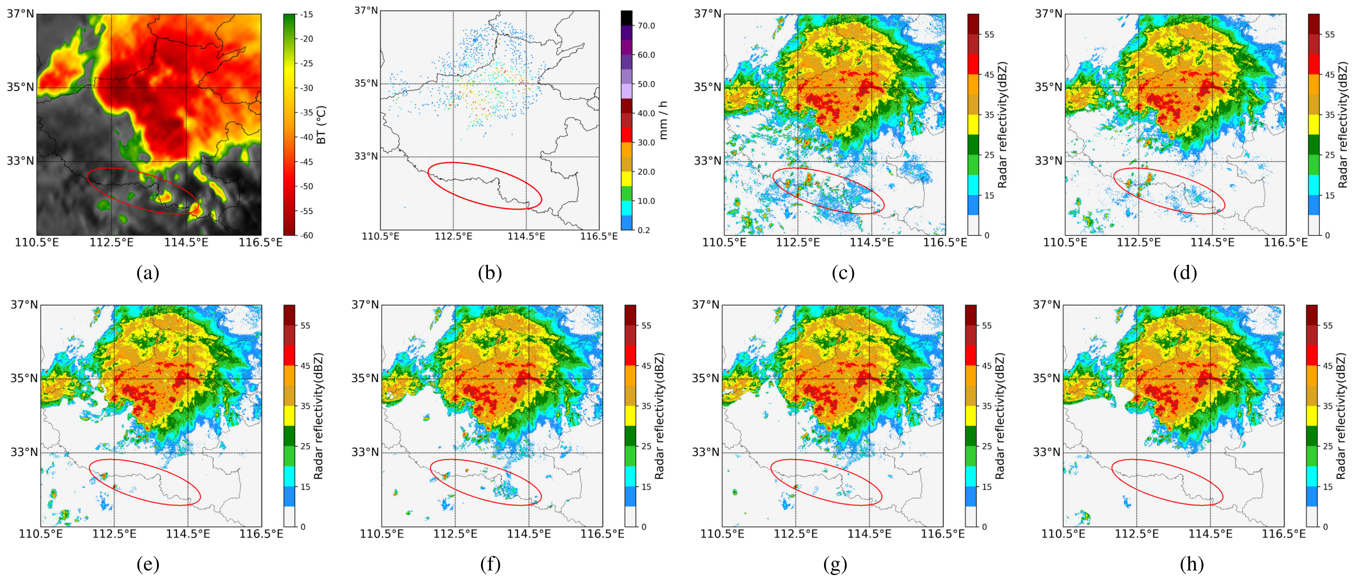

Fig. 8. Case study of Henan strong convective event at 03:00 UTC on July 20, 2021. (a) Satellite observation (10.80 μm). (b) Rain gauge observation. (c) RCR. (d) CREF. (e) U-Net. (f) ResNet. (g) RepViT. (h) RepNPE-Net.

TABLE VI

GENERALIZATION PERFORMANCE COMPARISON OF VARIOUS NETWORKS

	ACC	PREC	RECALL	F1	POD	FAR	CSI	HSS
U-Net	0.667	0.809	0.428	0.560	0.428	0.191	0.389	0.331
VGG	0.669	0.812	0.429	0.561	0.429	0.188	0.390	0.334
ResNet	0.673	0.782	0.470	0.587	0.470	0.218	0.415	0.343
RepVGG	0.648	0.700	0.504	0.586	0.504	0.300	0.415	0.294
RepViT	0.654	0.730	0.476	0.576	0.476	0.270	0.405	0.305
UniRepLKNet	0.674	**0.845**	0.416	0.557	0.416	**0.155**	0.386	0.343
InternImage	0.670	0.823	0.422	0.558	0.422	0.177	0.387	0.335
RepNPE-Net	**0.684**	0.778	**0.506**	**0.613**	**0.506**	0.222	**0.442**	**0.363**

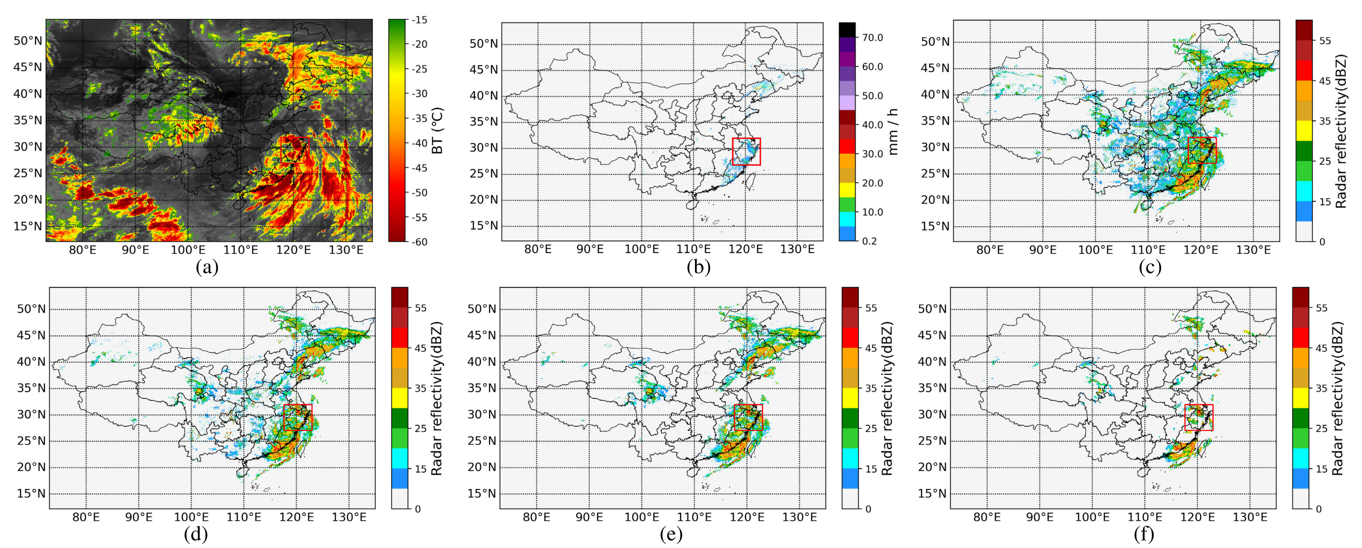

Fig. 9. NPE removal case of China region using various networks at 03:42 UTC on July 26, 2024. (a) Satellite observation (10.80 μm). (b) Rain gauge observation. (c) RCR. (d) CREF. (e) RepViT. (f) RepNPE-Net.

NPE-Net refers to the proposed network without reparameterization during the inference stage, while S-NPE-Net means that the standard residual convolution blocks of the proposed network are reparameterized for deployment. "AvgTime/Point" indicates the average inference time per rainfall point across the testing dataset, whereas "TotalTime (CN)" represents the total inference time required to generate classification results for the entire China region. From the table, it can be found that the introduction of the dual-stream convolution structure and PELA block increases the parameter count and computational

TABLE VII
COMPUTATIONAL COMPLEXITY OF VARIOUS NETWORKS

Network	Parameter ($\times 10^6$)	AvgTime/Point (s)	TotalTime (CN) (h)
U-Net	4.78	3.83	8.01
VGG	14.79	2.25	9.10
ResNet	11.19	2.66	8.62
RepVGG	**2.25**	3.08	6.95
RepViT	2.32	2.40	7.23
UniRepLKNet	13.85	2.69	7.63
InternImage	4.92	2.34	6.64
NPE-Net	9.29	2.78	8.24
S-NPE-Net	9.10	2.42	7.05
RepNPE-Net	9.05	**1.88**	**5.28**

time of NPE-Net to 9.29×10^6 and 2.78 s, respectively, compared to RepViT. Leveraging multibranch reparameterization, the mean computational time of S-NPE-Net is reduced to 2.42 s. Moreover, the computational efficiency of RepNPE-Net is further improved, with its inference time reduced to 1.88 s. Notably, the total computational time for the entire China region is reduced to 5.28 h. The results indicate that HCR reparameterization technology can merge multibranch and multiconvolution structures into a single equivalent convolution during inference, significantly accelerating computation while preserving recognition performance, demonstrating its superior applicability and efficiency in large-scale scenarios.

VI. CONCLUSION

In this study, we present a novel RepNPE-Net to identify and eliminate NPEs of radar reflectivity data from FY-4A satellite BT observations. The network incorporates two dual-stream convolution structure-based reparameterized modules, RepDCM and RepADCM, which effectively integrate standard and DS convolutions to enhance feature extraction and representation capabilities. Notably, the RepADCM module introduces a PELA block, enabling the network to focus on spatially significant positional features, thereby improving classification accuracy. Additionally, the proposed HCR technology facilitates multibranch and multiconvolution training while enabling inference with single equivalent convolutions, significantly reducing computational complexity while maintaining performance. Experimental results demonstrate that RepNPE-Net outperforms existing methods across various evaluation indices, achieving superior overall performance and generalization ability. Notably, time complexity tests further illustrate its superior inference efficiency across per-point basis and the whole China region, making it a robust solution for enhancing radar data quality and advancing meteorological applications.

Despite achieving improved performance, the study is currently limited to China (Shanghai and Shandong), which may introduce geographic biases when applied to regions with different climatic or meteorological conditions. In future works, we plan to incorporate global datasets from NOAA and ECMWF to further enhance the recognition capability and geographic generalization of the proposed method.

ACKNOWLEDGMENT

The authors gratefully acknowledge the support of NVIDIA Corporation for the donation of the GPU used for this research. Additionally, they sincerely thank the anonymous reviewers for their insightful feedback, which greatly contributed to the improvement of this work.

REFERENCES

[1] W. Li, H. Chen, L. Han, and W.-C. Lee, "StarNet: A deep learning model for enhancing polarimetric radar quantitative precipitation estimation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4106513.

[2] J. Si, H. Chen, and L. Han, "Enhancing weather radar reflectivity emulation from geostationary satellite data using dynamic residual convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4201711.

[3] X. Guo, L. Zhang, Y. Chen, L. Han, Y. Ge, and Y. Sha, "Using multi-source data to remove non-precipitation echoes in weather radar data," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 7689–7692.

[4] C. Cheng, L. Zhang, H. Liang, and L. Zhang, "Elimination method of non-precipitation echoes in weather radar based on Meteorological Satellite data," *Meteorol. Disaster Reduction Res.*, vol. 40, no. 2, pp. 141–145, 2017.

[5] X. Ding, Y. Guo, G. Ding, and J. Han, "ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1911–1920.

[6] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.

[7] A. Wang, H. Chen, Z. Lin, J. Han, and G. Ding, "Rep ViT: Revisiting mobile CNN from ViT perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15909–15920.

[8] X. Ding, X. Zhang, J. Han, and G. Ding, "Diverse branch block: Building a convolution as an inception-like unit," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10881–10890.

[9] A. Guo et al., "RLCFE-Net: A reparameterization large convolutional kernel feature extraction network for weed detection in multiple scenarios," *Expert Syst. Appl.*, vol. 274, May 2025, Art. no. 126941. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417425005639

[10] X. Ding et al., "UniRepLKNet: A universal perception large-kernel ConvNet for audio, video, point cloud, time-series and image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5513–5524.

[11] X. Luo, H. Zhang, Y. Xie, H. Zhang, X. Xu, and S. He, "L3Net: Localized and layered reparameterization for incremental learning," *Neural Netw.*, vol. 188, Aug. 2025, Art. no. 107420. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608025002990

[12] X. Wu, X. Zhang, Z. Xiao, L. Hu, R. Higashita, and J. Liu, "Expert-like reparameterization of heterogeneous pyramid receptive fields in efficient CNNs for fair medical image classification," 2025, *arXiv:2505.13039*.

[13] P. K. A. Vasu, J. Gabriel, J. Zhu, O. Tuzel, and A. Ranjan, "Mobileone: An improved one millisecond mobile backbone," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 7907–7917.

[14] M. Steiner and J. A. Smith, "Use of three-dimensional reflectivity structure for automated detection and removal of nonprecipitating echoes in radar data," *J. Atmos. Ocean. Technol.*, vol. 19, no. 5, pp. 673–686, May 2002.

[15] Y.-H. Cho, G. W. Lee, K.-E. Kim, and I. Zawadzki, "Identification and removal of ground echoes and anomalous propagation using the characteristics of radar echoes," *J. Atmos. Ocean. Technol.*, vol. 23, no. 9, pp. 1206–1222, Sep. 2006.

[16] L. Tang, J. Zhang, C. Langston, J. Krause, K. Howard, and V. Lakshmanan, "A physically based precipitation–nonprecipitation radar echo classifier using polarimetric and environmental data in a real-time national system," *Weather Forecasting*, vol. 29, no. 5, pp. 1106–1119, Oct. 2014.

[17] J. J. Gourley, P. Tabary, and J. P. Du Chatelet, "A fuzzy logic algorithm for the separation of precipitating from nonprecipitating echoes using polarimetric radar observations," *J. Atmos. Ocean. Technol.*, vol. 24, no. 8, pp. 1439–1451, Aug. 2007.

[18] J. M. Krause, "A simple algorithm to discriminate between meteorological and nonmeteorological radar echoes," *J. Atmos. Ocean. Technol.*, vol. 33, no. 9, pp. 1875–1885, Sep. 2016.

[19] V. Lakshmanan, C. Karstens, J. Krause, and L. Tang, "Quality control of weather radar data using polarimetric variables," *J. Atmos. Ocean. Technol.*, vol. 31, no. 6, pp. 1234–1249, Jun. 2014.

[20] H. Lee, J. C. Park, J. G. Kim, and S. Kim, "Line-shaped non-precipitation echo detection using fuzzy inference system," in *Proc. Joint IFSA World Congr. NAFIPS Annu. Meeting (IFSA/NAFIPS)*, Jun. 2013, pp. 1074–1079.

[21] M. Tahanout and J. P. D. Chatelet, "FILCOH—A novel technique to reduce ground clutter echoes in precipitation radars operating in multiple PRT," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1967–1985, Mar. 2021.

[22] P. Lepetit et al., "Using deep learning for restoration of precipitation echoes in radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5100914.

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, 2015, pp. 234–241.

[24] J. Guo et al., "Reparameterization convolutional neural networks for handling imbalanced datasets in solar panel fault classification," *Eng. Appl. Artif. Intell.*, vol. 150, Jun. 2025, Art. no. 110541. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095219762500541X

[25] X. Ding, H. Chen, X. Zhang, J. Han, and G. Ding, "RepMLPNet: Hierarchical vision MLP with re-parameterized locality," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 568–577.

[26] X. Ding et al., "ResRep: Lossless CNN pruning via decoupling remembering and forgetting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4490–4500.

[27] M. Mastyło, "Bilinear interpolation theorems and applications," *J. Funct. Anal.*, vol. 265, no. 2, pp. 185–207, Jul. 2013.

[28] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn.*, 2010, pp. 807–814.

[29] W. Xu and Y. Wan, "ELA: Efficient local attention for deep convolutional neural networks," 2024, *arXiv:2403.01123*.

[30] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31× 31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11953–11965.

[31] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2015, pp. 1–11.

[32] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[34] W. Wang et al., "InternImage: Exploring large-scale vision foundation models with deformable convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14408–14419.

[35] P. J. Roebber, "Visualizing multiple measures of forecast quality," *Weather Forecasting*, vol. 24, no. 2, pp. 601–608, Apr. 2009.

[36] C. A. Doswell, R. Davies-Jones, and D. L. Keller, "On summary measures of skill in rare event forecasting based on contingency tables," *Weather Forecasting*, vol. 5, no. 4, pp. 576–585, Dec. 1990.

Jianwei Si received the B.Sc. degree in safety science and engineering from Liaoning Technical University, Fuxin, China, in 2018, and the M.Sc. degree in computer science and technology from Qingdao University, Qingdao, China, in 2021. He is currently pursuing the Ph.D. degree in artificial intelligence with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao.

His research interests include atmospheric remote sensing, radar meteorology, image processing, and deep learning.

Lei Han (Member, IEEE) received the B.Sc. degree in engineering mechanics and the M.Sc. degree in automatic control from Harbin Engineering University, Harbin, China, in 1998 and 2001, respectively, and the Ph.D. degree in atmospheric remote sensing from Beijing University, Beijing, China, in 2008.

From 2001 to 2004, he was a Software Engineer with Lucent Research and Development, Qingdao, China. From 2013 to 2014, he was a Visiting Scholar with the Earth Observation Laboratory (EOL), National Center for Atmospheric Research (NCAR), Boulder, CO, USA. Currently, he is a Professor with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao. His research interests include data science and AI, remote sensing of natural disasters, and deep learning.

Lejian Zhang received the B.Sc. degree in information engineering from Nanjing Institute of Meteorology, Nanjing, China, in 2003, and the M.Sc. and Ph.D. degrees in atmospheric physics and atmospheric environment from Nanjing University of Information Science and Technology, Nanjing, in 2006 and 2010, respectively.

Currently, he is a Professor of engineering with the Meteorological Observation Center, China Meteorological Administration, Beijing, China. His research interests include weather radar data quality control and product algorithm development.

Chuanxin Li received the B.Sc. degree in electronic information engineering from the Ocean University of China, Qingdao, China, in 2023, where he is currently pursuing the M.Sc. degree in new generation electronic information technology (including quantum technology) with the Faculty of Information Science and Engineering.

His research interests include atmospheric remote sensing, image processing, and deep learning.